Diff 8 HW

1) $p=3,\ q=9$

2) many to one

3) $a=10$
$d=2$

$u_n = 10+2(n-1)$

$10+2(n-1)=50$
$n-1=20$
$n=21$

At 8th session $n_{lengths}=24$

1	15	24
2	20	26
3	25	28
4	30	30

$\therefore n=4$

$S_n=\frac{n}{2}(20+2(n-1)) \geq 400$

$n(9+n) \geq 400$
$n^2+9n-400 \geq 0$

$n=16$ or $n=-25$ but $n>0$
$\therefore n=16$

This assumes incorrect!

4 $3x+1=2x+3 \quad \therefore x=2$
or
$3x+1=-2x-3 \quad \therefore 5x=-4$
$x=-\frac{4}{5}$

graph it

$x=-4$

$x=-\frac{2}{3}$

on pos. side of ax.
so can only flip sign of graph where both graphs intersect.

5 $r\sin(\theta+a)= r\cos a\sin\theta + r\sin a\cos\theta$

$r\sin a=1$
$r\cos a=\sqrt{3}$

$\therefore \tan a=\frac{1}{\sqrt{3}} \therefore a=\frac{\pi}{6}$

$\therefore r\cdot\frac{1}{2}=1$
$r=2$

$\therefore \sqrt{3}\sin\theta+\cos\theta = 2\sin(\theta+\frac{\pi}{6})$

Stretch graph by scale factor 2 parallel to y axis and translate by $\begin{pmatrix}-\frac{\pi}{6}\\0\end{pmatrix}$

Consolidation

1 i) $t = 1 - x$

$y = (1-x)^2 - 4$

$= x^2 - 2x - 3$

ii) $t = \sqrt{\frac{x}{2}}$

$y = t^{-1}$

$= \left(\frac{x}{2}\right)^{-\frac{1}{2}}$

iii) $x^2 = 4\cos^2\theta + 4\sin\theta\cos\theta + \sin^2\theta$

$y^2 = \cos^2\theta - 4\sin\theta\cos\theta + 4\sin^2\theta$

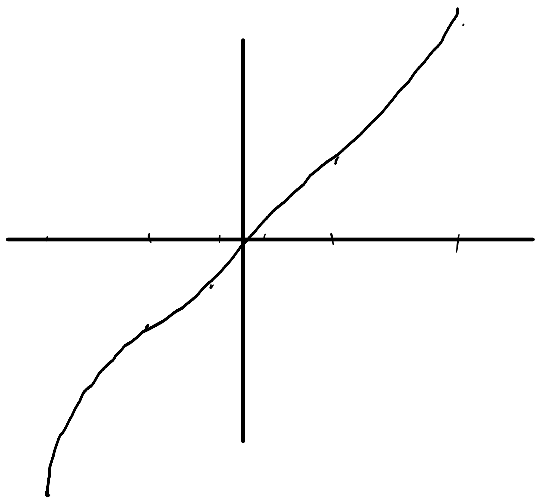
$x^2 + y^2 = 5\cos^2\theta + 5\sin^2\theta$

$= 5$

$x^2 + y^2 = 5$

2 i)

t	-3	-2	-1	0	1	2	3
x	9	4	1	0	1	4	9
y	-27	-8	-1	0	1	8	27



$t = x^{\frac{1}{2}}$

$y = \pm x^{\frac{3}{2}}$

3 i) $t = 0$

ii) $t = 1, \; t = -1$

$\therefore \; (0, -4)$ and $(0, 4)$

curve does not cross x axis.

iii) $\frac{y}{2}+x = t+\frac{1}{t}+t-\frac{1}{t}$

$= 2t$

$\frac{y+2x}{4} = t$

$x = \frac{y+2x}{4} + \frac{-4}{y+2x}$

$4x = y+2x+\frac{-16}{y+2x}$

$4x(y+2x) = (y+2x)^2 - 16$

$= y^2 + \cancel{4yx} + 4x^2 - 16$

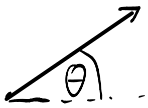
$\cancel{4yx + 8x^2 =}$

$y^2 = 4x^2+16$

$y = \sqrt{4x^2+16}$

$= 2\sqrt{x^2+4}, \quad x<0.$

4



i) $t = \frac{x}{16\cos\theta}$

$y = \frac{16\sin\theta}{16\cos\theta}x - 5t^2$

$= x\tan\theta - \frac{5x^2}{64\cos^2\theta}$

$= x\tan\theta - \frac{5}{64}x^2\sec^2\theta$ Note: $\sec^2\theta = 1+\tan^2\theta$

$-\frac{5}{64}x^2(1+\tan^2\theta$

$-\frac{5}{64}x^2 - \frac{5}{64}x^2\tan^2\theta$

$\therefore y = -\frac{5}{64}x^2\tan^2\theta + x\tan\theta - \frac{5}{64}x^2$

ii) Ball at ground when $y=0$

$16t\sin\theta - 5t^2 = 0$

$8t - 5t^2 = 0$

$t(8-5t) = 0$

$\therefore t = 0$ or $t = \frac{8}{5}$

$\rightarrow x = 16t\cos\theta$

$= 8\sqrt{3}\,t$

$= \frac{64\sqrt{3}}{5} = 22.2\text{m}$

iii) $y = 16t\sin\theta - 5t^2$

$= 8t - 5t^2$ (when $\theta = 30$)

$y' = 8 - 10t$

max y when $y' = 0$

$10t = 8$

$t = \frac{4}{5}$

$\rightarrow y = 3.2\text{m}$

5 i) When $t=1$:

$x=-1$

$y=6$

$\therefore (-1,6)$

ii) $\frac{dx}{dt}=3t^2-2t$

$\frac{dy}{dt}=10t-\frac{1}{t^2}$

$\frac{dy}{dt}\cdot\frac{dt}{dx}=\frac{10t^3-1}{t^2}\cdot\frac{1}{3t^2-2t}$

$=\frac{10t^3-1}{3t^4-2t^3}$

iii) $\left.\frac{dy}{dx}\right|_{t=1}=\frac{10-1}{3-2}=9$

iv) tangent: $y-6=9(x+1)$ normal: $y-6=-\frac{1}{9}(x+1)$

$y=9x+15$ $y=-\frac{1}{9}x+\frac{53}{9}$

6 At point $t=a$

$\frac{dx}{dt}=2t$

$\frac{dy}{dt}=2$

$\therefore \frac{dy}{dx}=t^{-1}$

$=a^{-1}$ when $t=a$

$\therefore y-2a=a^{-1}(x-a^2)$

$y=\frac{x}{a}+a$

when $y=0$, $\frac{x}{a}=-a$

$x=-a^2$ $(-a^2,0)=A$

When $x=0$, $y=a$ $(0,a)=B$

7i) $x=2\theta-2\sin\theta$

$\frac{dx}{d\theta}=2-2\cos\theta$

$y=2-2\cos\theta$

$\frac{dy}{d\theta}=2\sin\theta$

$\therefore \frac{dy}{dx}=\frac{2\sin\theta}{2-2\cos\theta}=\frac{\sin\theta}{1-\cos\theta}$

ii) When $\theta = \frac{\pi}{2}$, $x = \pi - 2$ and $y = 2$. $P = (\pi-2, 2)$.

$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{2}} = 1$

RTQ

$\therefore\ y - 2 = \cancel{x - \pi + 2}\ -(x - \pi + 2)$

$y = \cancel{x - \pi + 4}\ -x + \pi - 2$

$y = \pi - x$

At $y = 0$, $x = \underline{\pi - 4}$ $\quad A = (\pi-4, 0)$

iii) $\cancel{x = \pi - 4}$

$\cancel{(\pi - 4, 0)}\ (\pi, 4)$

<u>Exam Qs</u>

1) a) $x = t(1+t^3)^{-1}$

$y = t^2(1+t^3)^{-1}$

$\frac{dx}{dt} = t\left(-(1+t^3)^{-2} \cdot 3t^2\right) + (1+t^3)^{-1}$

$= -3t^3(1+t^3)^{-2} + (1+t^3)^{-1} \quad = -\frac{3t^3}{(1+t^3)^2} + \frac{1+t^3}{(1+t^3)^2}$

$= \frac{1-2t^3}{(1+t^3)^2}$

$\frac{dy}{dt} = -t^4(1+t^3)^{-2} + 2t(1+3t)^{-1}$

$\left.\frac{dy}{dx}\right|_{t=1} = \frac{\left.\frac{dy}{dt}\right|_{t=1}}{\left.\frac{dx}{dt}\right|_{t=1}}$

$= \frac{0.25}{-0.25} = \underline{-1}$

b) $x^3 = \frac{t^3}{(1+t^3)^3}$

$y^3 = \frac{t^6}{(1+t^3)^3}$

$xy = \frac{t^3}{(1+t^3)^2}$

$x^3 + y^3 = \frac{t^6 + t^3}{(1+t^3)^3}$

$= \frac{t^3(1+t^3)}{(1+t^3)^3}$

$= \frac{t^3}{(1+t^3)^2}$

$\therefore$ true

2 a) $\frac{dy}{d\theta} = 3\cos\theta$

$\frac{dx}{d\theta} = -2\sin\theta$

$\therefore \frac{dy}{dx} = \frac{3\cos\theta}{-2\sin\theta} = -\frac{3}{2}\cot\theta$

b) we know that $y-6 = m(x-2)$

$y = m(x-2)+6$

$3\sin\theta = -\frac{3}{2}\cot\theta(2\cos\theta - 2)+6$

$= -3\cot\theta(\cos\theta-1)+6$

$\sin\theta = -\cot\theta(\cos\theta-1)+2$

$\sin^2\theta = -\cos\theta(\cos\theta-1)+2\sin\theta$

$= -\cos^2\theta+\cos\theta+2\sin\theta$

$\sin^2\theta+\cos^2\theta = 2\sin\theta+\cos\theta$

$1 = 2\sin\theta+\cos\theta$

c) $2\sin\theta+\cos\theta=1$

$\theta=0$ or:

$2\sin\theta = (1-\cos\theta)$

$4\sin^2\theta = 1-2\cos\theta+\cos^2\theta$

$4(1-\cos^2\theta)$-

$4-4\cos^2\theta =$

$5\cos^2\theta - 2\cos\theta - 3 = 0$

$(5\cos\theta+3)(\cos\theta-1)=0$

$\therefore \underbrace{\cos\theta = -\frac{3}{5}}_{\theta=2.21}$ or $\underbrace{\cos\theta=1}_{\text{known}}$

$\therefore \theta = 0$ or 2.21